

Quick guides

Microtubule dynamic instability

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What are microtubules? One of three types of protein polymer that make up the cellular cytoskeleton, an elaborate array of filaments that are used to establish cell shape, facilitate cell motility, organize organelles and assist with cell division.

Microtubules are stiff tubes, about 25 nm in diameter. During interphase, they serve as tracks on which organelles and the nucleus are positioned by molecular motor proteins. During mitosis, microtubules form a structure called the mitotic spindle which physically segregates the chromosomes into the two daughter cells.

What is microtubule dynamic instability? Microtubules have special dynamic properties. In a population of microtubules, at any point in time, a subset of microtubules are rapidly growing while others are quickly shrinking (although sometimes some microtubules also sit still in a 'paused' state). This intriguing property is explained by the observation that individual microtubules switch randomly between growing and shrinking states, sometimes changing back and forth several times in the course of their lifetime. This combination of growth, shrinkage and rapid transitions between the two is known as dynamic instability.

Why is dynamic instability useful biologically? Speed. Dynamic instability allows the cell to rapidly reorganize the cytoskeleton when necessary. Dynamic microtubules are individually short-lived, so arrays of microtubules are continuously in the process of re-creation. Because microtubule

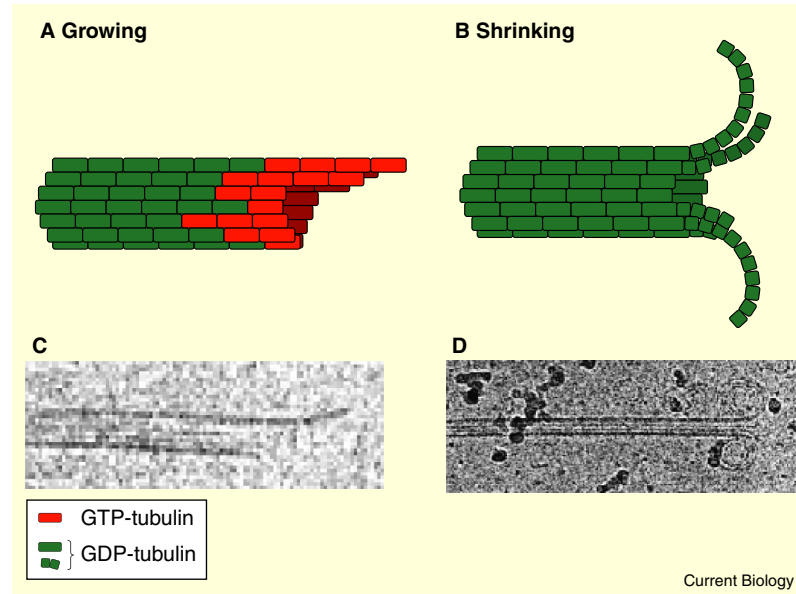


Figure 1. Microtubule ends in growing and shrinking states.

(A) Microtubules grow by addition of GTP-tubulin subunits (red). After they are incorporated into the microtubule, subunits are hydrolyzed to become GDP-tubulin (green). (B) GDP-tubulin is most stable in a curved state. Once microtubule shrinkage begins, possibly by loss of the GTP-tubulin at the end, the GDP-tubulin is free to splay out and the microtubule rapidly shrinks. (C,D) Cryo-electron micrograph images of microtubules in growing (C) and shrinking (D) states. (Images in (C) and (D) reproduced with permission from, respectively, Chretien *et al.* J Cell Biol. 129, 1311–1328 and Muller-Reichert *et al.* Proc. Natl. Acad. Sci. USA 95, 3661–3666.)

growth and shrinkage are active processes, consuming energy, this turnover can be fast, on the order of minutes. This means that an array of microtubules can adapt quickly to changes in the environment, adopting new spatial arrangements in response to cellular needs. Examples include the reorganization of the cytoskeleton in the transition to mitosis and the extension of growth cones from neurons.

How does dynamic instability work? Microtubules are made from subunits of the protein tubulin which are bound to the energy carrier guanosine triphosphate (GTP). The cell consumes energy to keep the concentration of GTP-tubulin high above the critical concentration for polymerization, far from equilibrium, so that subunits rapidly associate with microtubule ends and the microtubules grow.

After subunits are incorporated into a microtubule, their GTP is hydrolyzed into GDP, releasing energy. It is believed that some of this energy goes to deform the tubulin subunit, changing it so it

would be most stable in a curved state. This puts pressure on the microtubule to splay apart like a banana peeling.

However, the GDP-tubulin is not free to curve outwards while it is trapped in the microtubule lattice — the curving can only begin at the ends. While the ends are stable, a microtubule will grow, but once an end begins to come apart, the splaying propagates down the microtubule (Figure 1). The energy stored in the tubulin subunits is released as the microtubule rapidly shrinks.

What makes ends fall apart? We do not know what triggers ends to come apart. One possibility is that a layer of non-hydrolyzed, relatively stable GTP-tubulin remains at the microtubule end. In this case, the microtubule as a whole is stable and continues to grow. If this protective 'cap' is lost, the microtubule begins to splay. Alternatively, some other relatively rare 'tearing' event might begin microtubule shrinkage.

So if dynamic instability consumes energy, can it do

work? Yes! Growing microtubules can push against things, producing forces up to 4pN — comparable to the force produced by a kinesin motor, and enough to buckle the microtubules. Also, if the end of a shrinking microtubule is attached to an object, it can pull as it shrinks. In fact, microtubules attached to special sites on chromosomes are thought to use this type of force to pull the chromosomes apart during anaphase.

How is dynamic instability regulated? Some microtubule-associated proteins, or MAPs, can attach along the sides of the microtubules, slowing or reversing shrinkage. These MAPs are dominant in interphase cells and neurons, which is why microtubules in these cells are not very dynamic. Other MAPs, dominant in mitosis, attach to microtubule ends, where they can stabilize or destabilize microtubules by changing the frequencies of transitions between growing and shrinking states. The activity of many of these MAPs is highly regulated in time and in space.

Why is spatial regulation of dynamic instability important? Dynamic instability allows microtubules to rapidly explore space. If microtubules are preferentially stabilized or destabilized when they reach certain locations, the cell can build microtubule arrays with specific shapes. For example, during mitosis, selective stabilization of microtubules that encounter chromosomes is thought to help form the mitotic spindle.

Where can I find out more?

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Protein power strokes

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What is a power stroke?

A power stroke is large, rapid structural change in a protein that can be used to do mechanical work. A power stroke has a size on the order of the dimension of the protein itself (several nanometers); this distinguishes it from the much smaller localized structural changes that occur when a protein binds to a ligand or catalyzes a chemical reaction (the size of a chemical bond or a few Ångströms). The concept was first formulated for the protein myosin II which drives the contraction of muscle [1]. Subsequently, the term has been used to describe the conformational changes that take place in other motor proteins, such as the kinesins and the dyneins, as well as other macromolecular machines, such as protein translocases, the bacterial motor, ATP synthase and polymerases.

What drives a power stroke? The idea is that the protein has two different structural states, and that the binding or unbinding of a small molecule — or a chemical change in a bound molecule such as the hydrolysis of ATP — switches the protein from being primarily in structural state 1 to being primarily in structural state 2 (Figure 1A). If the structural states differ by a distance d , then the power stroke can do work against an external loading force F provided that $F \cdot d \leq \Delta G$, where ΔG is the decrease in free energy between the two chemical states. The power stroke of myosin II defined by crystallographic studies is illustrated in Figure 1B.

How is a power stroke measured? The first hint of the power stroke came from single-molecule recordings on myosin II interacting with an actin filament using an optical trap.

The recordings were difficult to interpret, however, because phosphate dissociates very shortly after the myosin binds to actin and so it was difficult to get a fix on the initial binding position (left-hand side of Figure 1B). But it was subsequently discovered that a different isoform, myosin I, undergoes a large conformational change upon ADP release: a rapid movement of 5 nm between the long-lived ADP and nucleotide-free states could therefore be well resolved [3]. The amplitude of this power stroke was confirmed by cryo-electron microscopy images of actin filaments ‘decorated’ with myosin I heads: addition of ADP to the nucleotide-free molecule induced a counter rotation of the lever arm domain. Such ADP-induced power strokes have subsequently been seen in several myosin-related proteins and kinesin.

How are the chemical and structural changes coupled?

To understand transitions between states, our static picture has to be replaced by one in which a protein fluctuates about an average or most-common structure, so the size of the power stroke will vary about a mean value from one observation to another. There are two ‘extreme’ views on how a chemical change could lead to a structural change. In the first, the chemical change strictly precedes the structural change: the changes in chemical bonds associated with a change in the ligand produce a highly strained conformation that relaxes into the final structural state. One can think of the chemical change cocking a spring that, when released, drives the protein forward. In the second, the structural change precedes the chemical one: the protein fluctuates, because of thermal forces, from its initial state to its final one before being trapped there by the chemical change. One can think of a door fluctuating into its open position against its hinge: once it has opened far enough, the chemical change